SCOPE OF ACCREDITATION TO ISO/IEC 17025:2017

BECKMAN COULTER, INC.
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CALIBRATION

Valid To: March 31, 2021 Certificate Number: 3436.01

In recognition of the successful completion of the A2LA evaluation process, accreditation is granted to this laboratory to perform the following calibrations[1, 4]:

I. Fluid Quantities

Parameter/Equipment	Range	CMC[2, 3] (±)	Comments
Airborne Particle Counter –			
Flow	(2.5 to 105) L/min	2.4 %	Volumetric flow meter, ± 2 %, TSI 4043/4143/4045
Particle Size	(0.3 to 25) µm	9.5 %	Thermo spheres ± 2.5 % for liquid particle ± 3.5 % for dry particle 10 µm and above
Hydraulic Particle Counter (PODS) –			
Flow	50 mL/min	2.8 %	100 mL graduated cylinder/stop watch
Count Accuracy	(4 to 6) µm(c) (6 to 21) µm(c)	10 % 18 %	ISO medium test dust
Particle Size	(42.3 to 71) µm	9.8 %	NIST soda lime glass spheres, ± 2.5 %

Parameter/Equipment	Range	CMC[2, 3] (±)	Comments
Hydraulic Particle Counter (8011) –			
Count Accuracy	(4 to 6) µm(c) (6 to 25) µm(c)	8.4 % 21 %	ISO medium test dust
Particle Size	(50.0 to 70.0) µm	7.4 %	NIST traceable spheres ± 3.5 %
Hydraulic Particle Counter (8012) –			
Count Accuracy	(4 to 6) µm(c) (6 to 25) µm(c)	12 % 22 %	ISO medium test dust
Particle Size	(50.0 to 70.0) µm	7.2 %	NIST traceable spheres ± 3.5 %
Pharma Liquid Counter/System –			
Flow	(10 to 25) mL/min (25 to 50) mL/min	2.4 % 4.8 %	Stopwatch and balance
Particle Size	(1.25 to 289) µm	5.6 %	Thermo spheres ± 2.5 %
HRLD Sensor for Water –			
Flow	(10 to 20) mL/min (20 to 200) mL/min	6.8 % 4.6 %	Rotometer ± 3 %
Particle Size	(1.25 to 289) µm	8.0 %	Thermo spheres ± 2.5 %

[1] This laboratory offers commercial calibration service.

[2] Calibration and Measurement Capability Uncertainty (CMC) is the smallest uncertainty of measurement that a laboratory can achieve within its scope of accreditation when performing more or less routine calibrations of nearly ideal measurement standards or nearly ideal measuring equipment. Calibration and Measurement Capabilities represent expanded uncertainties expressed at approximately the 95 % level of confidence, usually using a coverage factor of $k = 2$. The actual measurement uncertainty of a specific calibration performed by the laboratory may be greater than the CMC due to the behavior of the customer's device and to influences from the circumstances of the specific calibration.

[3] In the statement of CMC percent is defined as the percentage of the reading.

[4] This scope meets A2LA's *P112 Flexible Scope Policy*.

Accredited Laboratory

A2LA has accredited

BECKMAN COULTER, INC.

Loveland, CO

for technical competence in the field of

Calibration

This laboratory is accredited in accordance with the recognized International Standard ISO/IEC 17025:2017 *General requirements for the competence of testing and calibration laboratories*. This laboratory also meets R205 – *Specific Requirements: Calibration Laboratory Accreditation Program*. This accreditation demonstrates technical competence for a defined scope and the operation of a laboratory quality management system (*refer to joint ISO-ILAC-IAF Communiqué dated April 2017*).

Presented this 5th day of April 2019.

Vice President, Accreditation Services
For the Accreditation Council
Certificate Number 3436.01
Valid to March 31, 2021

For the calibrations to which this accreditation applies, please refer to the laboratory's Calibration Scope of Accreditation.